

Synthetic Fair-Value Strategy

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1. Introduction

A prediction-market contract is a binary claim that pays \$1.00 if an outcome occurs and \$0.00 if it does not; before settlement it trades between those bounds at whatever price buyers and sellers agree reflects the outcome's probability. Polymarket US launched regulated, real-money contracts of this kind in January 2026, and its coverage of the 2026 FIFA World Cup listed, for each tracked player in each match, three separate contracts on a central limit order book: at least one goal, at least one assist, and at least one goal-or-assist. Each has its own order book and its own update cadence.

These three contracts are not independent instruments. Scoring a goal is one of the ways to record a goal-or-assist, so the compound contract can never be worth less than either single-stat leg:

$$P(\text{goal} \cup \text{assist}) \geq \max(P(\text{goal}), P(\text{assist})).$$

When the compound contract is quoted less actively than the legs that decompose it, its price drifts below that floor. This is a pricing error the market makes against itself, observable without reference to any position taken. Across 157,595 simultaneously captured three-leg quotes, the compound bid falls below the goals bid in 28.81% of pre-kickoff observations, and the compound contract carries a $3.81\times$ wider spread and rests $5.6\times$ less size than the goals leg it can be priced against.

Whether that staleness reverts is a separate question, and it is tested rather than assumed. Conditioning on positions entered while the compound contract was cheap relative to its formula fair value, the mispricing narrows toward fair value 63.0% of the time, against 15.1% under a volatility-matched random walk ($z = 6.94$, $p < 0.000001$). A second strategy trades confirmed public events directly: when a goal, a substitution, or a final whistle makes one side of a contract near-certain, a resting order on the other side has not yet requoted, and whoever acts before the book adjusts is trading a latency gap.

The purpose of this paper is to measure the pricing inefficiency, test whether it is a consistent and statistically significant effect, and evaluate whether either strategy built around it beats the market's own pricing rather than merely trading faster than it. Only the first does. The speed strategy is profitable, but because it buys contracts already priced near certainty, its win rate is the arithmetic of those prices and not a measurable edge — a distinction Section 5 makes precise.

2. Data Pipeline and Screening Architecture

The analysis relies on a fully automated data pipeline that identifies, validates, and tracks pre-kickoff prop markets while minimizing contamination from fundamentally driven repricing. Its objective is to construct a clean sample of simultaneously observed three-leg quotes and to measure price behavior in the window where the compound contract de-stales.

The pipeline runs unattended on Windows Task Scheduler and integrates three data sources. First, market and event data is read from Polymarket US's public gateway API, which carries a live score and match clock on each event object. Second, the account's own fills, balances, and settlements are read from the authenticated portfolio API, signed with an Ed25519 key over the message `{timestamp}{method}{path}`. Third, event timestamps used for latency measurement are drawn from neutral match reports (FIFA, ESPN, Al Jazeera), never from Polymarket's own price data, so that the latency figures are not circular.

Polling is tiered by time to kickoff: every twenty minutes beyond six hours out, tightening to every sixty seconds inside the final thirty minutes. Market discovery is structural, not a hand-maintained list. The set of soccer leagues is pulled live from the platform's own sports endpoint, which returned seventy leagues besides the World Cup as of 2026-08-08; each league's games are then enumerated directly against the API.

A clean three-leg observation is recorded only when the goals, assists, and goals-or-assists contracts for the same player can all be read from the same pre-kickoff market state in a single pass. Pairing all three legs at identical capture timestamps yields 157,595 exactly-simultaneous observations, of which 124,571 fall in the forty-eight hours before kickoff. This is the sample used throughout the paper.

Every fill the account made is reconciled at ingestion: each trade row's quantity is checked against its cost divided by its price, and the local trade log is cross-checked one-to-one by trade identifier against a fully paginated pull of the account's complete live history (392 trades). The trading client used for these reads exposes only balance, position, and activity lookups; it contains no order-placement methods.

3. Signal Identification

For a given player in a given match, let g , a , and ga be the traded prices of the “1+ goals,” “1+ assists,” and “1+ goals-or-assists” contracts. Under a statistical-independence approximation, inclusion–exclusion gives a fair value for the compound contract:

$$P(\text{goal} \cup \text{assist}) = P(\text{goal}) + P(\text{assist}) - P(\text{goal} \cap \text{assist}) \approx g + a - g a.$$

The signal is the difference between that fair value and the contract's actual price:

$$\text{gap} = (g + a - g a) - ga.$$

A positive *gap* means the compound contract is trading cheap relative to its own legs, which is a buy signal. Across 208 clean pre-kickoff samples, *gap* averaged +0.0106 and exceeded 0.05 in absolute value in 31 of 208 cases — a meaningful mispricing on an instrument priced between \$0.00 and \$1.00.

The reference constraint is the logical floor stated in Section 1: because a goal is one route to a goal-or-assist, a correctly priced compound contract can never trade below either single-stat leg. A violation of that inequality is a pricing error that requires no model to identify — only three simultaneous quotes.

This is not riskless arbitrage. The independence approximation carries model risk in both directions: a goal is negatively correlated with the same player's own assist, which pushes true $P(\text{goal} \cap \text{assist})$ below $g a$, while a player heavily involved in attack has elevated odds of both

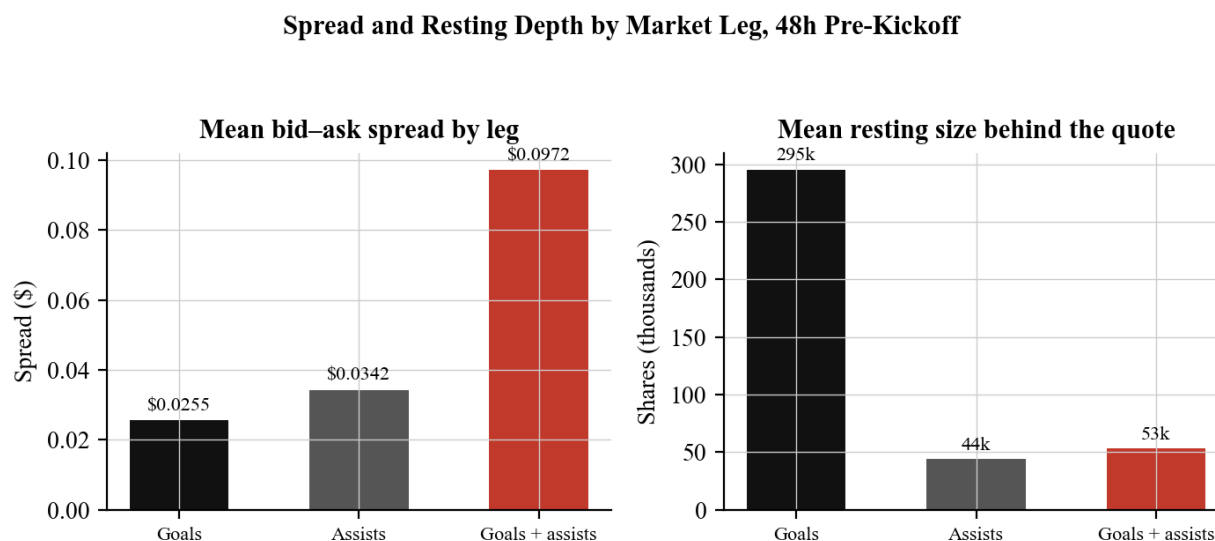
outcomes for reasons the formula cannot see. The strategy is therefore statistical relative-value arbitrage with adverse-selection exposure: a resting bid improved toward fair value can still be filled by a counterparty more informed than the formula.

4. Data Analysis

Every snapshot of quotes and book depth is bucketed by time to kickoff. The four figures below test, directly against the data, the mechanisms the two strategies depend on.

Spread and Resting Depth by Market Leg

The plot below displays the mean bid–ask spread and the mean resting size for each of the three legs, measured over the forty-eight hours before kickoff for the fourteen highest-coverage players.

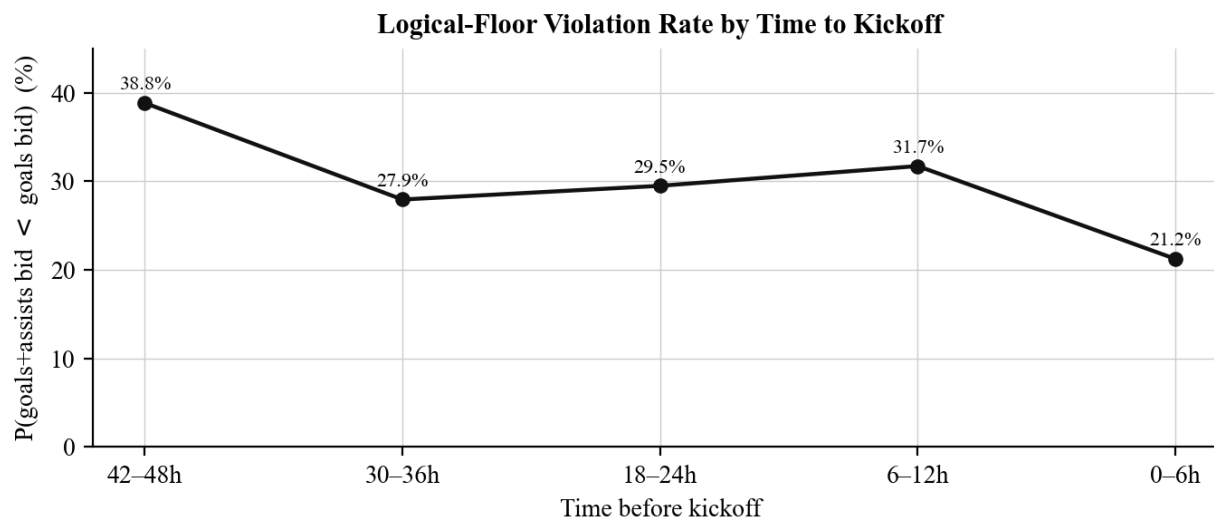


The plot shows that the compound contract is structurally the stale one, on both measures at once. It carries a $3.81\times$ wider spread than the goals leg (\$0.0972 against \$0.0255) and rests $5.6\times$ less size behind its quotes (52,966 against 295,156 shares), widening to $7.5\times$ in the final four hours. The goals leg is the anchor: its spread is essentially flat near \$0.019 for the entire two days, and its book grows from roughly 295,000 to 376,000 shares as retail attention arrives. The compound market behaves nothing like it — it opens at a \$0.157 spread and compresses to \$0.047 by kickoff,

but its resting size never grows at all. Retail flow arrives into the simple market and largely ignores the compound one.

Logical-Floor Violation Rate by Time to Kickoff

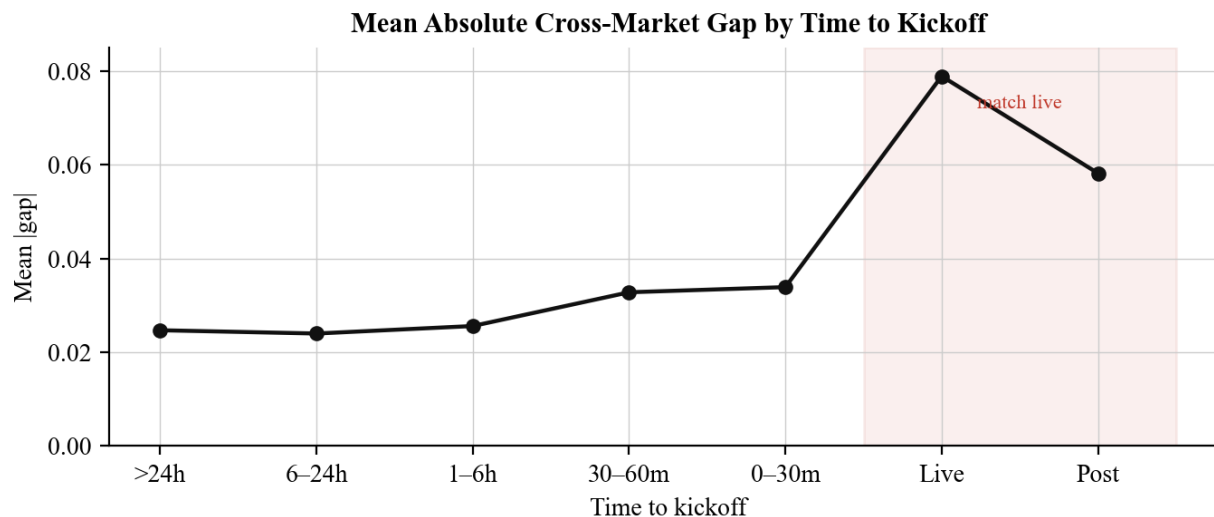
The plot below displays the fraction of pre-kickoff observations in which the compound bid sat below the goals bid, grouped by hours before kickoff.



The plot shows that the best resting bid on the strictly-more-likely outcome sat below the best resting bid on a subset of it in 28.81% of pre-kickoff observations (35,894 of 124,571), by a mean of \$0.088. The violation is highest far from kickoff — 38.83% at forty-two to forty-eight hours out — and decays to 21.23% inside the final six hours, exactly as a de-staling process predicts. It also runs opposite to fame: Lionel Messi's compound book violated the identity six times in 7,287 observations, while players one tier below the household names — famous enough for their markets to attract flow, not so famous that the compound leg was watched as closely as the simple one — carried violation rates above 20%.

Mean Absolute Cross-Market Gap by Time to Kickoff

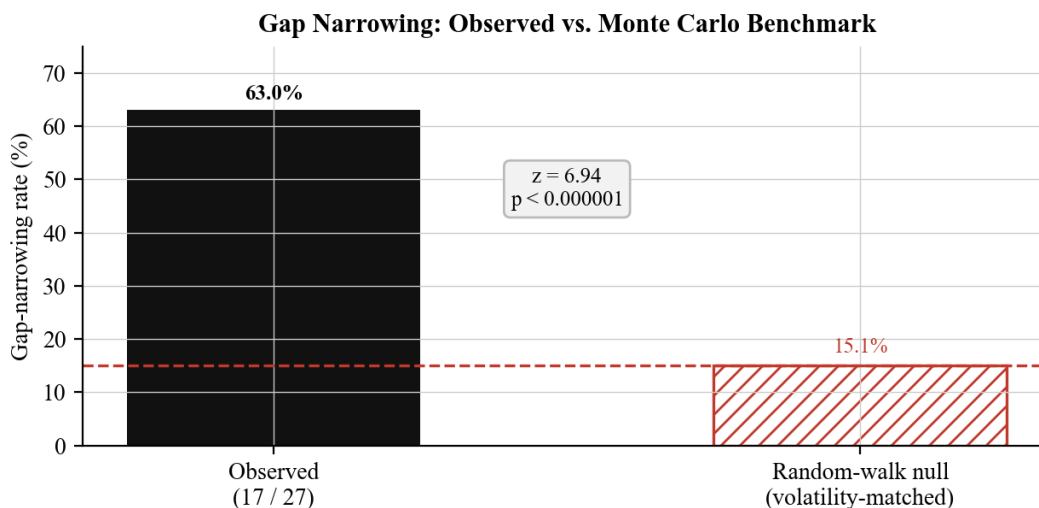
The plot below displays the mean absolute difference between the compound price and its formula fair value, by time bucket, from more than a day before kickoff through live play.



The plot shows that the gap is roughly flat near 0.025 for the entire pre-kickoff window and then more than triples, to 0.079, once the match goes live. Real scoring events hit the three legs at different speeds, and the resulting lag is the inefficiency the second strategy trades.

Gap Narrowing: Observed vs. Monte Carlo Benchmark

To test whether the pre-kickoff mispricing actually reverts rather than merely drifting, per-step increments of gap were computed from 182,760 consecutive same-player, same-game observations (median step about sixty-six seconds), giving a mean of approximately zero and a standard deviation of $\sigma = 0.0227$ per step. For each of the twenty-seven cleanly pre-kickoff-exited positions with a resolvable gap reading at both entry and exit, 5,000 random-walk paths were simulated — each starting at that position's actual entry gap and taking its actual number of observed steps — and the fraction of paths ending with a smaller absolute gap than they started with was recorded.



The plot shows that the observed data narrows far more often than the volatility-matched benchmark: 63.0% of the real positions closed toward fair value, against 15.1% of the simulated paths. Because the simulation is calibrated directly from the real second-by-second movements and assumes no drift, the difference reflects structure in the price process rather than a parameter choice.

5. Statistical Testing

Each test below is a one-sample proportion z -test. Where the null probability is an entry price rather than a fixed benchmark, the test asks whether the strategy's realized results differ from what the market's own pricing implied — the correct question for a strategy that may simply be fast rather than better informed.

Test 1: Is the gap mean-reversion real?

The null hypothesis is that the gap signal follows a driftless, volatility-matched random walk, which narrows 15.1% of the time.

$$H_0 : p = 0.151 \quad H_1 : p > 0.151 \quad n = 27, \quad \hat{p} = 0.630$$

$$z = \frac{0.630 - 0.151}{\sqrt{\frac{0.151(1 - 0.151)}{27}}} = 6.94, \quad p < 0.000001.$$

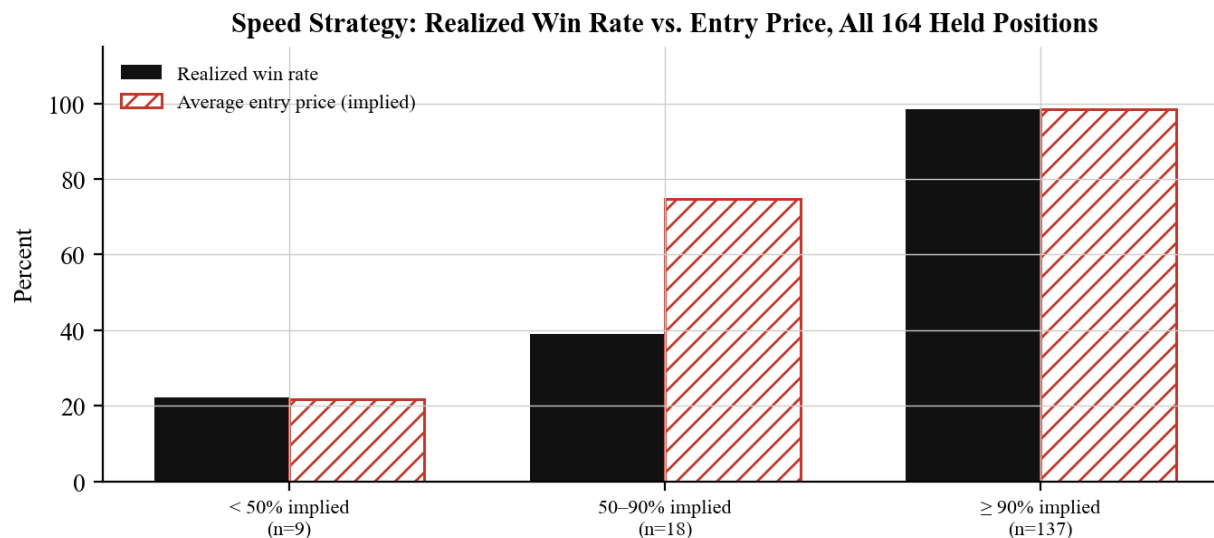
Because the p -value is effectively zero, we reject the null hypothesis. The one caveat worth stating: contract prices live in $[0, 1]$, so gap is bounded while the simulated walk is not, and a bounded process starting far from zero must revert eventually simply because it cannot escape. Two things bound how much that can explain. The effect size is large — 63.0% against 15.1% is not a margin a boundary artifact plausibly manufactures at these gap magnitudes — and the low simulated rate is driven by the walk having room to drift over a typical holding period, not by the walk being pushed outward. Re-running the null with reflecting barriers at the price bounds would raise the simulated rate somewhat but would not close the gap.

Test 2: Is the speed strategy calibrated?

The speed strategy trades a confirmed public event before the resting book requotes. Which side it ends up holding depends on the event: a confirmed goal or assist leaves it long “Yes,” a confirmed substitution or full-time whistle leaves it long “No.” Both are the same trade — act on public news faster than the book adjusts — applied to opposite events, so they are tested together. The null hypothesis is that the market was already pricing those contracts efficiently, so the true win probability equals the average price the account paid for the side it held.

$$H_0 : p = 0.917 \quad H_1 : p \neq 0.917 \quad n = 164, \quad \hat{p} = 0.878$$

$$z = \frac{0.878 - 0.917}{\sqrt{\frac{0.917(1 - 0.917)}{164}}} = -1.81, \quad p = 0.070.$$



We fail to reject the null hypothesis, and the shape of the result matters more than the failure. The 137 positions entered above 90% implied confidence resolved the account's way 98.5% of the time against a 98.5% implied rate — calibrated to the decimal, and uninformative for exactly that reason: buying a contract the market has priced at 98% and watching it resolve at 98% is arithmetic, not skill. A perfect record in that band carries little weight; across the 48-position “Yes” sub-bucket alone, a flawless outcome has probability $0.987^{48} \approx 0.53$ under the market's own pricing. The 18 positions entered between 50% and 90% implied resolved at 38.9% against a 74.7% implied rate — a real shortfall, but one that two positions dominate (Section 6), leaving only a thin low-confidence tail behind the effect.

The test is also underpowered against the edge the mechanism actually claims. At $n = 164$ and a null of 91.7%, the realized rate would have had to clear roughly 95% to reject at conventional significance, while Section 7 claims only a two-to-three-point timing edge. The correct reading of $p = 0.070$ is “no edge this sample can detect, plus a small miss concentrated in the least-priced entries” — not a demonstrated edge in either direction. The speed strategy's profit is real and mechanically reliable; it is simply not something a win rate can attribute to skill rather than to buying contracts the market had already priced near certainty.

What the win rates do and do not show

The speed strategy's headline win rate — 87.8% across its 164 held positions — invites a reading it cannot support. The strategy buys contracts the market has already priced between roughly 80% and 99% and holds them to settlement, so a high win rate is the arithmetic those entry prices predict, not a result. A perfectly calibrated market handed the same trades would produce the same number. The only question that separates edge from arithmetic is whether the realized rate exceeds the price the account paid, and Test 2 answers it: the realized rate sits just below the price paid, not above it, and what miss there is concentrates in a thin band of low-confidence entries rather than reflecting a pricing edge. That is not the profile of a strategy that prices better than the market.

That leaves the speed strategy's profitability resting entirely on a timing claim — that it buys after a goal, substitution, or whistle is public but before the resting book has requoted. A win rate cannot establish this, because a stale price and a fair price at 99% resolve the same way. The only direct evidence is the reaction-time sample in Section 2: seven independently timestamped goals, a mean latency near 91 seconds, every entry landing while the contract still traded below the post-event fact. Seven observations is a genuine distribution but a thin one, and it is the load-bearing evidence for the entire speed strategy. The patience strategy is the only one in this paper whose edge survives a test against the market's own pricing.

Multiple comparisons and in-sample validity

Bonferroni-correcting the two tests above (threshold $0.05/2 = 0.025$) changes no conclusion: the mean-reversion test clears it by orders of magnitude, and the speed-strategy calibration test does not reach even the uncorrected 0.05. The larger limitation is that the gap signal was designed and evaluated on the same thirty-five-day dataset, with no out-of-sample holdout. A complete census of every soccer competition Polymarket US lists finds that none of the other seventy leagues currently carries the three-parallel-market structure the signal requires, so a genuine replication needs a second tournament that does not yet exist. The collector already discovers events across

all seventy leagues, so that data begins accumulating the moment one lists a player goal or assist market again.

6. Risk

A key risk to the patience strategy arises when the compound contract's staleness does not close before kickoff. The exit is a passive resting sell, not a market order — it depends on a counterparty's order arriving — and when it does not fill in time the position is carried into live play, where match variance layers on top of an already-late exit. Of 193 round-trip positions, only 56 fully exited before kickoff as intended; the remaining 137 were carried into the match. The 56 clean exits won 96.4% of the time; the 137 delayed positions won only 80.3%, with lower average returns — a real, quantified cost of fill risk, though the clean-exit figure is partly a selection artifact, since a position moving the wrong way is more likely to still be open at kickoff than sold at a loss beforehand.

Jeremy Doku's goals-or-assists position ahead of Belgium–Senegal is a clear example. It was bought across three pre-kickoff fills at \$0.29 to \$0.31, a blended \$0.2978, and was not sold before kickoff. Carried into the match, it filled across six tranches at \$0.38 to \$0.41, realizing +\$25.18 (+27.6%) and +\$23.57 on its two largest tranches. The mispricing it was entered on was real; it simply closed late, and a first move the other way would have produced the opposite result on the same delay.

The speed strategy carries a different risk on its “No” positions: a confirmed substitution or full-time whistle caps a player's remaining probability, but two positions were entered well short of certainty and resolved against the call. Charles De Ketelaere (goals-or-assists, Spain–Belgium, 07-10), entered at an average “No” price near 79%, lost \$73.47 — the largest single-position loss in the strategy data. Jovo Lukić (goals, Bosnia and Herzegovina–Qatar, 06-24), entered near 88%, lost \$46.38. Both are genuine signal misses rather than execution artifacts; they are the two largest dollar losses in the strategy and most of the 50–90% bucket's calibration gap in Test 2. Above 90% implied — 137 of the strategy's 164 held positions — realized and implied rates match to within a point.

To keep the latency measurement honest, event timestamps are taken only from neutral match reports, never from Polymarket's price data, and second-half goals are excluded rather than approximated: converting a second-half minute to a wall-clock time requires the real halftime-break length, and assuming a standard fifteen minutes can swing a computed latency by 300 seconds — more than the entire quantity being measured.

7. Execution Model

Pre-Trade Identification and Filtering

- **Clean three-leg observation.** A candidate surfaces only when all three legs quote from the same pre-kickoff market state and the gap exceeds a threshold; $|gap| > 0.05$ held on 31 of 208 clean samples.
- **Star-player universe.** Positions concentrate on high-coverage names — Messi, Mbappé, Haaland, Kane and similar — whose compound markets de-stale reliably enough to be exited into before kickoff. An obscure player can carry the same theoretical gap with far less confidence it closes and far less volume to exit into.

Entry Logic

- **The fair-value ceiling.** Rather than crossing the spread, the patience strategy repeatedly improves its own resting bid in small increments, capped strictly below the formula-implied fair value, so the position buys at or below its own fair-value estimate even in the best case for a counterparty. On the fills, 96.2% of pre-kickoff compound buys were priced below the formula value computed from the concurrent leg asks.
- **Passive only.** Section 4 shows the inefficiency sat almost entirely on the bid side; a crossable, locked arbitrage — buying the compound at its ask and selling the goals leg at its bid — existed in only 0.05% of observations. An aggressive strategy would have found nothing to take.
- **The confirmed-event trigger.** The speed strategy does not use price level as a trigger. On a confirmed goal, assist, substitution, or final whistle for a tracked player, the near-certain side is bought immediately at whatever price is showing and held to \$1.00 settlement.

Risk and Position Management

- **Fixed-dollar sizing.** Positions are sized as a fixed fraction of available capital rather than a share count, for consistency across contract prices.
- **Hard cap.** Total portfolio risk is capped at 5% of capital, checked at kickoff; a position built past that fraction across multiple fills is cut back with a below-market sell.
- **Breadth over size.** The speed strategy spreads capital across every still-unresolved candidate at the moment of the trigger rather than concentrating on one, since the confirming event makes each outcome close to certain rather than merely likely.

Exit Strategy

- **Staged sell-off.** Inside the final five minutes before kickoff, the patience strategy sells roughly a quarter of whatever remains once per minute, crossing the spread each round to guarantee that tranche fills.
- **Unconditional backstop.** A final sweep at kickoff itself closes anything the schedule leaves behind, so the position is always flat by kickoff — the property the trade record found missing in 137 of 193 cases.
- **Hold to settlement.** Once a speed-strategy entry lands near \$0.98 to \$0.99, the remaining margin is too thin, net of the trading fee, to be worth an active sell. Zero of 193 patience-strategy round-trips were ever sold at a price at or above \$0.90; every position that reached that band was held to settlement.

The live order-placement layer for the patience strategy was developed jointly with a collaborator on a separate account and is described here as reconstructed from that logic and cross-checked against the surviving trade ledger; every figure in this paper derives from the author's own account and its own records.

8. Final Weights

Two execution schedules follow from the model above. The pre-kickoff exit, which exists to close the fill-risk gap in Section 6, sells a fixed fraction of the remaining position each minute

inside the final five, with an unconditional backstop at kickoff:

Minutes to kickoff	Action	Fraction of remainder	Cumulative % closed
5	Staged sell	25%	25.0%
4	Staged sell	25%	43.8%
3	Staged sell	25%	57.8%
2	Staged sell	25%	68.4%
1	Staged sell	25%	76.3%
0	Unconditional sweep	100%	100%

Entries are not scheduled the same way — they are placed as the signal presents itself across the full pre-kickoff window. Of 86 pre-kickoff compound buys, the median fill landed 8.7 hours before kickoff (mean 14.6 hours), and the distribution has two human-scale clusters rather than a flat profile:

Window before kickoff	Share of pre-kickoff compound buys
18–24 hours	20%
6–18 hours	37%
Final 6 hours	43%

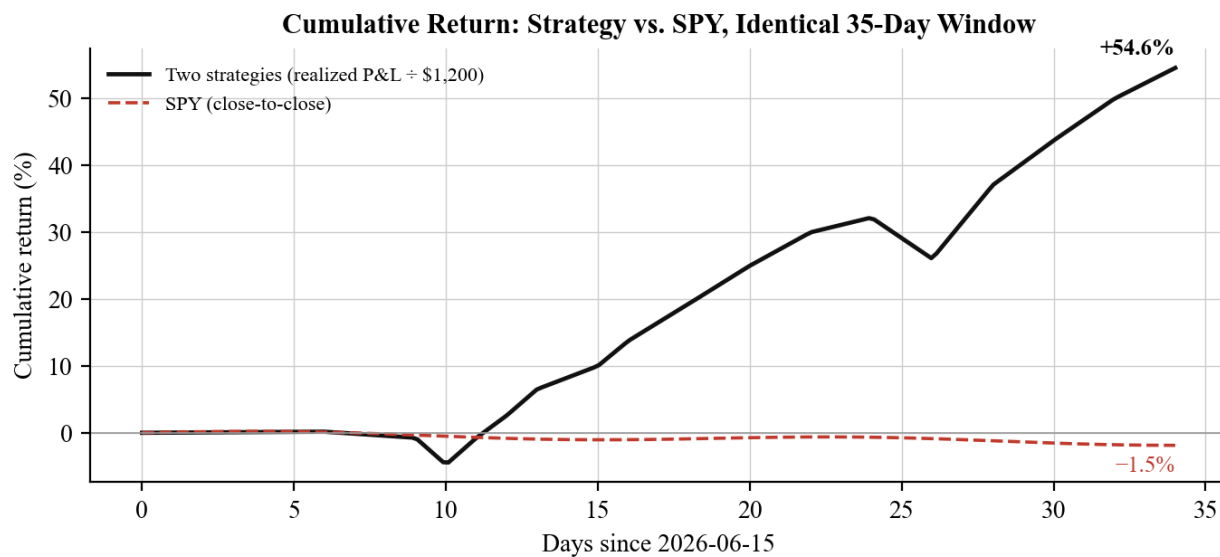
9. Results

The strategies were run against Polymarket US's World Cup player-prop markets over a thirty-five-day window, 2026-06-15 to 2026-07-19. Net capital contributed was \$1,200 across four completed deposits; the account carries zero open positions at close, so every figure reported is fully realized. Total realized profit over the window was \$671.99, and it has two parts.

The two strategies this paper analyzes account for \$655.27: \$304.64 from the patience strategy (compound positions actively sold, 193 closes) and \$350.63 from the speed strategy (confirmed-event trades held to settlement — \$319.71 on the 164 positions carried to the final whistle, plus \$30.92 on 16 closed by an active buy-to-cover beforehand). The other \$16.72 is the

net result of discretionary trades the account placed in seven bet types outside either strategy — match futures, advancement, match-winner, spreads, exact score, player shots, and shots on target, 26 markets across 89 fills, none of them player goal-or-assist props. These were not produced by either strategy's entry logic, are not analyzed further, and are excluded from every test and figure below.

Against SPY over the identical calendar window — real closes \$754.83 on 2026-06-15 to \$743.29 on the last trading day before the window ended — the two analyzed strategies returned +54.6% on contributed capital (\$655.27 on \$1,200), against SPY's −1.5%. The comparison establishes only that the return is not an artifact of a broad-market tailwind.



The two strategies behave differently, and only one carries a statistically validated edge:

	Patience (sold)	Speed strategy (held)
Positions	193	164
Win rate	85.0%	87.8%
Net P&L	+\$304.64	+\$319.71
Validated edge?	Yes ($p < 0.000001$)	No ($p = 0.070$)

The speed row covers the 164 positions held to settlement, on which the calibration test is run; the strategy realized a further \$30.92 on 16 positions closed by buy-to-cover beforehand,

for a total of \$350.63 as broken out above. The win-rate row is reported for completeness but discriminates almost nothing for the speed strategy: it bought contracts the market had already priced between 80% and 99%, so a high win rate is what those entry prices predict rather than a result (Section 5). The “validated edge” row is the one that separates the two, and only the patience strategy clears it.

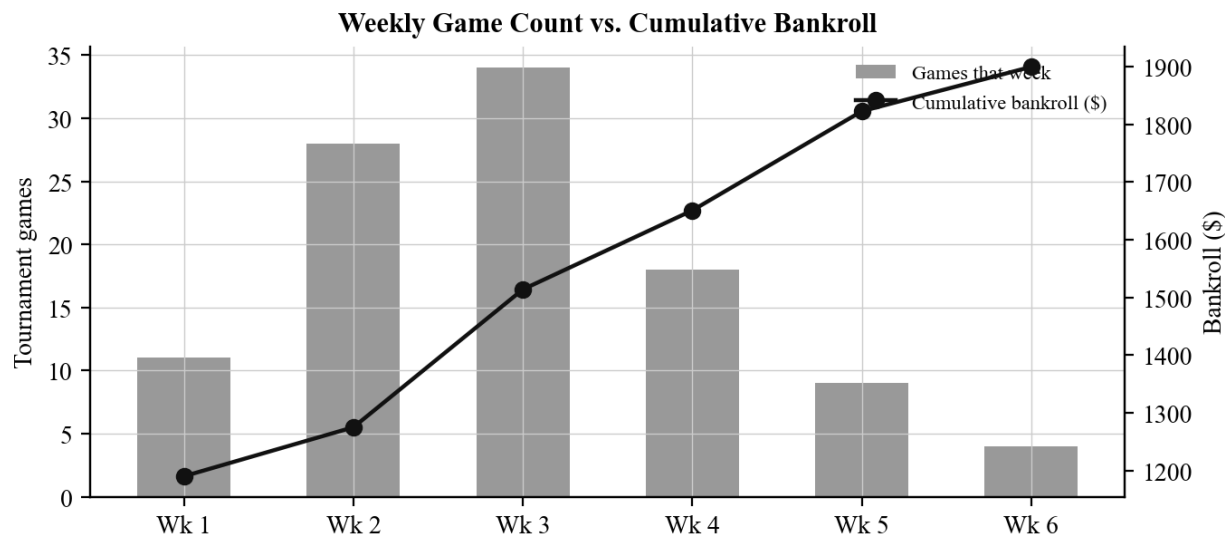
Within the patience strategy, the fill-risk split is the more informative comparison: the 56 positions that exited before kickoff won 96.4% of the time, and the 137 carried into live play won 80.3%.

Metric	Strategy	Benchmark (SPY)
Cumulative return, 35 days	+54.6%	−1.5%
Sharpe ratio, daily ($n = 31$)	0.42	—
Maximum drawdown	\$73.94 (11.9% of peak)	—

The annualized Sharpe is deliberately not reported. Thirty-one daily observations from a single bounded event put a wide confidence interval on the daily estimate, and scaling that interval by $\sqrt{365}$ produces a range broad enough to be uninformative while presenting as precision. Restricting the calculation to the period after all \$1,200 was deposited gives a daily Sharpe of 0.57 on $n = 23$ — higher, not lower, once the low-capital ramp-up is removed.

Capacity

The clearest single fact in the data about whether the approach generalizes is that capital grew exactly as opportunity shrank.



Tournament game count fell from 11 to 28 to 34 to 18 to 9 to 4 per week as the bracket narrowed, while the account's bankroll rose monotonically from roughly \$1,190 to \$1,823 across those same weeks. The account held the least capital when opportunity was greatest and the most when it was smallest. Combined with measured top-of-book depth of 1,406 to 9,762 shares against an average position size of \$17.98, and a median thirty-eight-minute lag between a game-ending event and settlement freeing the capital, the strategy is capacity-bound by construction.

10. Conclusion

This study demonstrates that a thinly-quoted compound prediction-market contract, priced against the more actively quoted single-stat contracts that decompose it, carries a quantifiable and mechanical pricing inefficiency. Its bid falls below the logical floor $P(\text{goal} \cup \text{assist}) \geq P(\text{goal})$ in 28.81% of 124,571 pre-kickoff observations, and that staleness reverts toward fair value far more often than a volatility-matched random walk predicts — 63.0% against 15.1%, $p < 0.000001$. This is the only edge in the paper that survives a test against the market's own pricing. A second strategy, trading confirmed public events directly, is profitable but its results are not evidence of skill: it buys contracts already priced at 80% to 99% and holds them, so its high win rate is arithmetic those prices predict, and a calibration test cannot distinguish it from its own entry pricing (87.8% realized against 91.7% implied, $p = 0.070$), with what miss there is confined to a thin tail of low-

confidence entries. What profit it makes rests on a timing claim the win rate cannot test and that only seven directly measured event latencies support. The account converted \$1,200 into \$671.99 in fully realized profit over thirty-five days — \$655.27 of it from the two strategies analyzed here, \$16.72 from discretionary trades outside them — outperforming SPY (+54.6% against −1.5% on the strategy figure) over the identical window.

The inefficiency is real, but it is capacity-bound by construction. The account held the least capital when tournament opportunity was greatest and the most when it was smallest, and measured book depth against realized position size leaves no room to scale without redesigning the execution layer. Ultimately, the findings suggest that a retail-heavy prediction market during a marquee event prices its simple contracts efficiently and its compound contracts against itself — an asymmetry a faster, better-modeled counterparty can capture at small size, but not, on this evidence, at institutional scale.